

Sample Images

HACM-10
Source
10 Classes

1,000 per class

Degradation Conditions

Motion Blur
Partial Occlusion
Low Illumination
Background Clutter

Ambiguity Conditions

Condition Combinations
0 – 4 Stacked

3 Annotators Label +
Confidence 0 – 100

Independent Rating

Response-Time Filter
Invariant-Response Filter
Label-Entropy Filter

Quality Control

27,456 Annotations
9,984 Images

Human Signals

μ_H | σ_H | H_label

Dataset Construction

Data Split

55% Train | 15% Validation | 15% Test | 15% Calibration

F1 F2 F3 F4 F5
F1 F2 F3 F4 F5
F1 F2 F3 F4 F5
F1 F2 F3 F4 F5
F1 F2 F3 F4 F5

VALIDATION TRAINING

5 Fold Cross-Validation

ResNet-18
MobileNetV2
ViT-B/16

Baseline Models

HA-ECN —Proposed Dual-Backbone Model

ResNet-50

Conv 7×7 s2 → Max Pool 3×3 → Residual Block ×16 → $7^2 \times 2048$

EfficientNet-B4

Conv 3×3 s2 → SE Block → MBConv Block ×32 → $7^2 \times 1792$

Loss: CMP + Human-Informed Cross-Entropy (λ = 0.5)

Softmax Average
0.5 / 0.5

20 Model Instances
4×5 folds

Model Architecture & Training

3.19× vs 2.55× |HACM|
0.134–0.138
8.4× human vs model
$\beta = 0.078$, $p = 8.8 \times 10^{-165}$

Evaluation Metrics

HACM (x)
p_AI – μ_H

Temperature Scaling

ECE 0.147 → 0.009

RAPS Conformal
Not Informative

→Feature for misalignment predictor

MC Dropout
T = 50

→Feature for misalignment predictor

Calibration & Analysis

Calibration Cliff → Architecture Sensitivity → Misalignment Predictor → Statistical Validation